T3000 API & Protocol

TrendLog Chart X-Axis Design · built Nov 13, 2026

TrendLog Chart X-Axis Design

Overview

The TrendLog chart X-axis uses **Solution 3: Hybrid Right Edge Rounding**. The right edge is always rounded *forward* to the nearest clean boundary based on the active timebase, then the left edge is computed by subtracting the window duration from that rounded right edge.

Rounding Rules

Under 1 hour (5m, 10m, 30m)

Right edge rounds **up to the next minute ending in 0 or 5**.

Now	Right edge
07:22	07:25
07:25	07:25 *(exact boundary, no shift)*
07:25:01	07:30
07:17	07:20

1 hour and above (1h, 4h, 12h, 1d, 4d)

Right edge rounds **up to the next full :00 hour**.

Now	Right edge
07:22	08:00
08:00	08:00 *(exact boundary, no shift)*
08:00:01	09:00

Custom range

No rounding. The user supplies an explicit start and end time; those values are used as-is.

Window Calculation

```
TEXT

rightEdge  = roundRight(currentTime, timebase)   // applies rules above
leftEdge   = rightEdge - windowDuration(timebase)
```

Timebase	Window duration	Tick step
5m	5 min	1 min
10m	10 min	1 min
30m	30 min	5 min
1h	60 min	10 min
4h	4 h	30 min
12h	12 h	1 h
1d	24 h	2 h
4d	4 days	12 h

Examples (now = 07:22)

Timebase	Left edge	Right edge	Ticks (sample)
5m	07:20	07:25	07:20, 21, 22, 23, 24, 07:25
10m	07:15	07:25	07:15, 16 … 07:25
30m	06:55	07:25	06:55, 07:00, 05, 10, 15, 20, 07:25
1h	07:00	08:00	07:00, 10, 20, 30, 40, 50, 08:00
4h	04:00	08:00	04:00, 04:30 … 08:00
12h	Apr28 20:00	Apr29 08:00	20:00, 21:00 … 08:00
1d	Apr28 08:00	Apr29 08:00	08:00, 10:00 … 08:00
4d	Apr25 08:00	Apr29 08:00	Apr25 08:00, 12:00 …

Live Mode (Auto-Scroll)

When the chart is in **live mode** (`isRealTime = true` , `timeOffset = 0`):

- The right edge is recalculated every second from `Date.now()` .
- When `now` crosses a rounding boundary (e.g. 07:25:00 → 07:25:01) the right edge automatically advances to the next boundary (07:30), and the whole window shifts forward.
- The red dashed **"now" line** marks the exact current time within the window. The small area to the right of the "now" line is the forward tail (at most one step width).

Scroll Behaviour (Left / Right arrows)

Action	Effect
← **Left**	Exits live mode (if active), shifts window back by exactly one timebase period. Each additional click shifts another full period back.
→ **Right**	Shifts window forward by one timebase period. When `timeOffset` returns to 0, live mode is automatically restored.
Disabled for	Custom date range

Zoom Behaviour (Zoom In / Zoom Out)

Zoom changes the active timebase without affecting the live/historical state.

Progression (shortest → longest):

```
TEXT
```

```
5m  →  10m  →  30m  →  1h  →  4h  →  12h  →  1d  →  4d
```

- **Zoom In (↑):** moves one step left in the progression (finer detail).
- **Zoom Out (↓):** moves one step right (wider view).
- Disabled at the ends of the progression and for Custom ranges.

Reset

Clicking **Reset** sets `timebase = 5m` , `timeOffset = 0` , and restores live mode.

Key Tick Label Rules

- Ticks are placed at clean multiples of the tick step within the window.
- On day-crossing timebases (12h, 1d, 4d), whenever the date changes the label shows `MonDD HH:MM`; other labels show only `HH:MM`.
- The right-edge tick label is always shown in **green** to distinguish the rounded boundary from data ticks.
- The exact "now" position is marked with a **red dashed vertical line** when live mode is active.